Midterm Exam

Time: 75 minutes

Name:

Student #:

- Answer each question in the space provided or on the back of a page with an indication of where to find the answer.
- Write your name and student number in the space provided on this sheet. Write your student number only in the space provided on other sheets of this paper.
- This is a closed book examination. One page cheat sheet is allowed
- If you find a question to be ambiguous or feel that you must make assumptions in order to complete your answer successfully, please clearly state those assumptions in your answer on the exam paper

Q1	
Q2	
Q3	
Q4	
Total	

Question 1 (MCQ):

Please select ALL the correct answers from the choices below.

1. Which of the following statements regarding dropout in neural networks is true?
   a. It is only used during training ☑
   b. It is used to increase the training speed
   c. The probability of dropping is a hyperparameter ☑
   d. Dropout is applied differently during training and testing. ☑
2. What is the role of the activation function in a neural network?
   a. Introduce non-linearity ☑
   b. Reduce the dimensionality of data
   c. Increase model complexity
   d. Apply regularization
3. What was the main benefit of using 1x1 convolutions in Inception modules?
   a. Reduce dimensionality ☑
   b. Introduce non-linearity
   c. Increase spatial resolution
   d. Apply batch normalization
4. What makes InternImage architecture distinct?
   a. It uses dynamic convolution kernels ☑
   b. Captures global and local dependencies more effectively ☑
   c. Relies on self-attention mechanisms like transformers
   d. Shallow architecture with fewer layers
5. Which of the following adjustments might improve the performance of an underfitting MLP model?
   a. Increase the number of hidden layers ☑
   b. Reduce the number of neurons in hidden layers
   c. Reduce the amount of regularization ☑
   d. Apply dropout with a high rate
6. What is the effect of initializing all weights in an MLP to the same value?
   a. The model will train normally
   b. The neurons in each layer will learn the same features ☑
   c. It helps with faster convergence
   d. It prevents overfitting
7. What happens if the predicted probability in cross-entropy loss is 1 and the actual label is 0?
   a. The loss is minimized
   b. The loss approaches zero
   c. The loss is undefined
   d. The loss approaches infinity ☑
8. Why is Mean Squared Error (MSE) not a suitable loss function for classification problems?
   a. MSE does not penalize incorrect classifications effectively ☑
   b. MSE does not provide a probabilistic output like cross-entropy loss ☑
   c. MSE leads to poor gradient updates for classification models
   d. MSE does not allow backpropagation in deep learning models.
9. In a one versus all multi-class classifier, which of the following is true
   a. We train C two-class classifiers first, and then use a fusion rule to fuse the results ☑
   b. Learn all sets of weights simultaneously ☑
   c. The method guarantees perfect separation between all classes
   d. One-vs-all requires fewer training examples than one-vs-one
10. When using cross-entropy loss with Softmax in a multi-class classification, what is a key requirement?
    a. The model outputs must sum to 1 ☑
    b. The output layer must be linear
    c. The input features must be normalized
    d. The loss function should be mean squared error
11. Which of the following statements is true regarding Stochastic Gradient Descent (SGD) compared to Full Batch Gradient Descent (BGD)?
    a. SGD updates weights after every training example, while BGD after processing entire dataset ☑
    b. BGD typically converges faster than SGD in non-convex optimization problems
    c. Using a decaying learning rate can improve SGD convergence stability ☑
    d. BGD guarantees convergence to the global minimum in all cases

Question 2 (Debugging)

(1) You are training a two-layer MLP with ReLU activations, and you notice that some neurons never activate during training and stop learning. After further analysis you discover that the input to the ReLu is always negative. What is the name of this problem and how could you modify the network to prevent this issue?

This issue is known as the **dying ReLU problem**, where some neurons output zero for all inputs and stop learning because their gradients become zero. This typically happens when the input to the ReLU activation function is always negative, leading to zero output and no gradient updates during backpropagation.

To prevent this issue, you can try the following solutions:

1. **Use Leaky ReLU or Parametric ReLU (PReLU)**
2. **Use better weight initialization:**
3. **Lower the learning rate**
4. **Use batch normalization**

(2) You are training a deep neural network and using sigmoid function as your non-linearity. You notice that the activations in the later layers are either all very large or all very close to zero. Your friend tells you that this has to do with the weight initialization. Do you agree with this statement? How would you solve this issue to improve the network's performance?

Your friend is correct—**improper weight initialization** can cause activations in later layers to become **either too large** or **too close to zero**, leading to the **vanishing gradient problem**. This occurs because the **sigmoid activation function** squashes inputs into the range **(0,1)**, making large positive inputs approach **1** and large negative inputs approach **0**. This causes gradients to shrink exponentially as they propagate backward through layers, slowing down learning.

Use Xavier (Glorot) Initialization for Sigmoid

Use Batch Normalization

Switch to a Different Activation Function

(3) At the beginning of training a deep neural network, you are trying to train to 100% training accuracy on a small sample of training data (~5-10 minibatches). However, you notice that the loss explodes to inf or NaN. One team member suggests that this has to do with your learning rate, while another suggests that this has to do with weight initialization. Which one is correct (or both or neither)? and explain what is happening and how to fix it.

Both are correct. L**earning rate** and **weight initialization** could be responsible for the exploding loss (`inf` or `NaN`)

1) **Reduce the learning rate: If using SGD, try decreasing it (e.g., from 0.1 → 0.01 → 0.001).**
2) **Use proper weight initialization:**

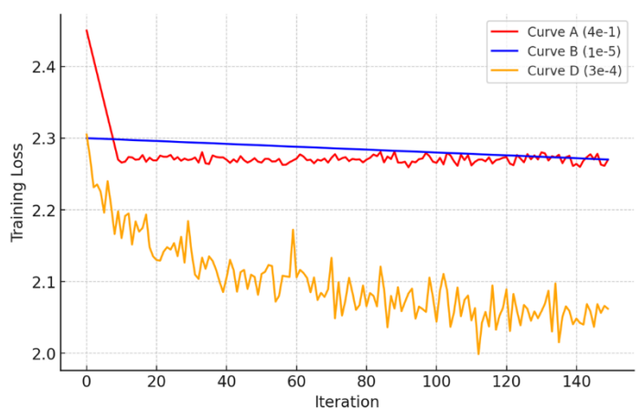
- For **ReLU/Leaky ReLU**: Use **He Initialization**
- For **Sigmoid/Tanh**: Use **Xavier/Glorot Initialization**

(4) When you are training your new neural network model, you used stochastic gradient descent (SGD) with momentum and L2 weight regularization while testing three different learning rates. The resulting training loss curves are displayed below, but you did not record which curve corresponds to which curve. The learning rates are in creasing order:

$$learning_rates = \{0.00001, 0.0003, 0.4\}$$

Your job is figure which learning rate belongs to which curve and justify your answer

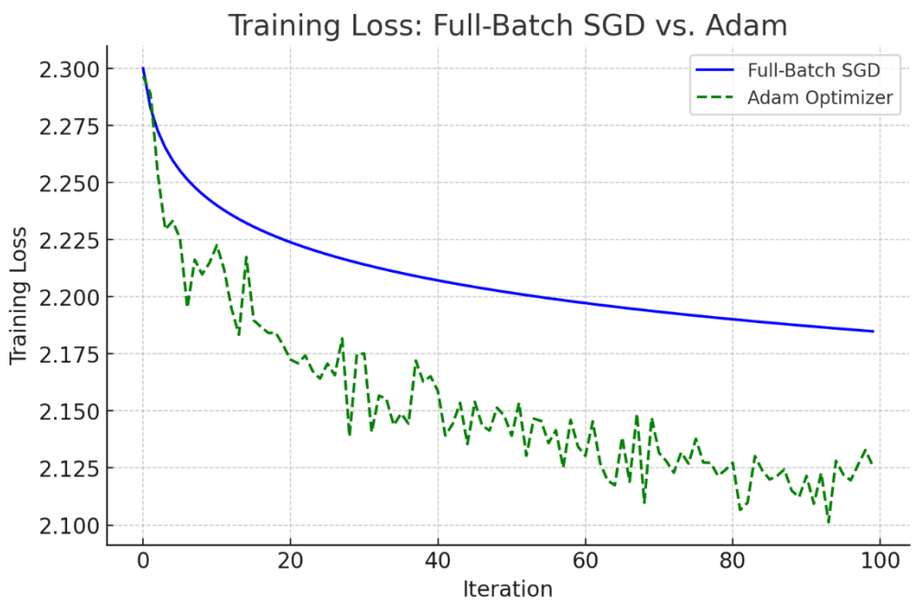
The answer is below:
Red curve A has a learning rate of 4e-1
Blue curve B has a learning rate of 1e-5
Orange curve C has a learning rate of 3e-4



(5) You are training a feed-forward neural network model for a regression problem. You produced the following curve of training loss over iterations.You used full batch gradient descent and Adam. Which figure belongs to which optimizer and why?

The answer is below:

Blue curve X is Full Batch SGD
Green curve Y is Adam Optimizer

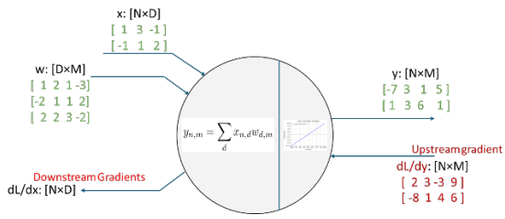


Question 3 (Theory)

(1) In the figure below list any **TWO** CNN networks that we studies in this course. Mention in 1-2 sentences the contribution that was proposed in this architecture

See Class notes. Any two should work

(2) One way to calculate the downstream gradient in the image below is to explicitly find the Jacobians $\frac{\partial y}{\partial x}$ and $\frac{\partial y}{\partial w}$. Explain what the downside of this explicit method of Jacobian calculations is and suggest a solution.



1. High Computational Cost & Memory Usage

2 Slow Computation in Deep Networks

3. Inefficiency in High-Dimensional Spaces

Solution: Use Reverse-Mode Autodiff (Backpropagation) or what we called in class implicit Jacobian calculations

Question 4 (Design and Coding)

(1) Consider the following dataset below with input features x_1 and x_2, Each datapoint in the table has a target value y.

Datapoint	x_1	x_2	y
p_1	1	3	2
p_2	3	2	3
p_3	2	2	1
p_4	1	1	4

(a) Write an equation for a linear regression model that fits this dataset

(b) Write a python code to find the wights of this model using normal equation (direct solution).

$$y = w_0 + w_1 x_1 + w_2 x_2$$

(b)

```
import numpy as np

X = np.array([
   [1, 1, 3],  # Bias term, x1, x2
   [1, 3, 2],
   [1, 2, 2],
   [1, 1, 1]
])
Y = np.array([2, 3, 1, 4])
W = np.linalg.inv(X.T @ X) @ X.T @ Y
print("Computed Weights:", W)
```

these values not required

I have computed the **multiple linear regression coefficients** (θ) using the normal equation. The result are:

- **Bias (Intercept)** w_0 = **4.818**
- **Weight for** x_1 w_1 = **-0.182**
- **Weight for** x_2 w_2 = **-1.000**

Final Regression Equation:

$$y = 4.818 - 0.182x_1 - 1.000x_2$$

This equation defines a **plane** in a **3D space** where x_1 and x_2 are the independent variables.

(2) [4] Assume you have a classification problem with 3 classes, and you come up with a CNN classifier. For each layer, calculate the number of weights, number of biases and the size of the associated feature maps. The notation follows this convention:

- CONV-K-N denotes a convolutional layer with N filters, each of them is of size K x K, Padding and stride parameters are always 0 and 1 respectively.
- POOL-K indicates a K x K pooling layer with stride K and padding 0.
- FC-N stands for a fully-connected layer with N neurons.

Layer	Activation map dimensions	Number of weights	Number of biases
INPUT	128 × 128 × 3	0	0
CONV-9-32			
POOL-2			
CONV-5-64			
POOL-2			
CONV-5-64			
POOL-2			
FC-128			
FC-3			

See the Table below

	Layer	Activation Map Dimer	Number of Weights	Number of Biases
1	INPUT	128 × 128 × 3	0	0
2	CONV-9-32	120 × 120 × 32	7776	32
3	POOL-2	60 × 60 × 32	0	0
4	CONV-5-64	56 × 56 × 64	51200	64
5	POOL-2	28 × 28 × 64	0	0
6	CONV-5-64	24 × 24 × 64	102400	64
7	POOL-2	12 × 12 × 64	0	0
8	FC-128	128	1179648	128
9	FC-3	3	384	3